INCH-POUND

MIL-T-50356A
19 March 1993
SUPERSEDING
MIL-T-50356(MU)
8 October 1969

MILITARY SPECIFICATION

THRUSTER, CARTRIDGE ACTUATED, M1A2 REMANUFACTURED METAL PARTS ASSEMBLY

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for the M1A2 Cartridge Actuated Thruster Remanufactured Metal Parts Assembly (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-A-2550 Ammunition, General Specification For

STANDARDS

MILITARY

MIL-STD-105 Sampling Procedures And Tables For Inspection By Attributes
MIL-STD-109 Quality Assurance Terms And Definitions
MIL-STD-413 Visual Inspection Guide for Elastomeric O-rings

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Standardization Branch (Code 8420), 101 Strauss Avenue, Indian Head, MD 20640-5035, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by sending a letter.

AMSC N/A FSC 1377

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:22Z
Check the source to verify that this is the current version before use.

(Unless otherwise indicated, copies of federal and military specifications and standards are available from Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following Government drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation (see 6.2).

DRAWINGS

Naval Surface Warfare Center, Indian Head (CAGE Codes 19200 and 19203)

8593855	Thruster, Cartridge Actuated, M1A2, Metal Parts Assembly
8797631	Box, Fiberboard (Inner), Packing, Ammunition, For Thruster, Cartridge Actuated, M1A2
8797632	Box, Packing, Ammunition, For Thruster, Cartridge Actuated, M1A2
9221261	Box, Fiberboard (Outer), Alternative, Packing Ammunition, for Thruster, Cartridge Actuated, M1A2

U.S. Air Force

TPO 00-891-5488	Thruster, Cartridge Actuated, M1A2

(Application for copies should be addressed to: Commander, Indian Head Division, Naval Surface Warfare Center, Attn: Data Control Branch (Code 8410), Indian Head, MD 20640-5035.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. All materials used in the manufacture of the thrusters shall conform to the specifications referred to on the respective drawings unless specific approval in writing covering a departure therefrom has been obtained from the cognizant Navy design activity prior to manufacture. When alternate materials or methods of manufacture are specified on the drawings, the bidder's selections shall be clearly stated in the proposal. The remanufactured thruster metal parts assembly shall contain a new shear pin, new O-rings, and a new piston locking spring.

3.2 Hydrostatic pressure. The body and head shall withstand a hydrostatic pressure of 7,500 pounds per square inch (psi) for 15 seconds, minimum, without leaking, deformation, or mechanical failure (see 6.3.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:22Z
Check the source to verify that this is the current version before use.

3.3 Protective finish. Protective finish shall be in accordance with the applicable specifications and drawings and shall apply to all surfaces of the component regardless of configuration. Remanufactured exterior components (see 6.4) shall have the existing protective finish stripped and the specified protective finish reapplied.

3.4 Residual magnetism. The metal parts assembly shall not deflect the indicator of the compass more than five degrees in either direction using the procedures of 4.5.3.

3.5 Shear pin strength. The shear pin, after application of protective finish, shall shear when a load of 48 ± 4 pounds is applied at the steady rate of 0.125 inch per minute.

3.6 Metal parts assembly. Parts shall be combined to produce an assembly which conforms to the requirements of the metal parts assembly drawing. All threaded joints shall be assembled with sufficient torque to bottom components.

3.6.1 Integrated plant products. When remanufacturing and loading of the assembly is performed in an integrated plant, the metal parts assembly requirement may be waived and the residual magnetism requirement performed on the loaded assembly only.

3.6.2 Threaded component parts. Class 2A or 2B thread gage shall be used on remanufactured component parts inspection versus class 3A or 3B thread gage required on new hardware for thrusters.

3.6.3 Piston locking spring and end sleeve. The piston locking spring shall enter the spring hole in the piston without any bending along its entire length. If axial misalignment of the piston locking spring occurs, the cap screw may be backed off sufficiently to permit alignment.

3.6.4 Trunnion rotation. Upon completion of assembly, the trunnion shall be free to rotate 360 degrees.

3.6.5 End sleeve rotation. Upon completion of assembly, the end sleeve shall be free to rotate 360 degrees in a clockwise direction.

3.6.6 Reclaimed parts. Remanufactured thruster parts which may be reclaimed shall meet the following requirements:

a. Piston. Only pistons with 1/2 inch length from back face to O-ring groove and 5/8 inch length from front face to back face of key escape groove shall be reclaimed. Nonconforming pistons which have the unlocking undercut of 9/16 inch length shall be modified to meet the 5/8 inch length requirement of Drawing 8593867. Touch-up plating to the machined area is acceptable.

b. Head. Only heads with internal bore diameter of 0.337 + 0.009 inch and no setscrew holes shall be reclaimed.

c. Body. Only bodies suitable for nylon pellet type thread fastening shall be reclaimed. The nylon pellet shall be located 0.775 ± 0.015 inch from end of body. The chamfer on

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:22Z
Check the source to verify that this is the current version before use.

the outer edge of the threaded end shall be 1/16 x 5/64 and the inner edge shall be 1/16 x 3/64. The length of the trunnion flat shall be 17/32.

d. Cartridge retainer. Only retainers suitable for nylon pellet type thread fastening shall be reclaimed. The nylon pellet shall be located 0.453 ± 0.015 inch from the end. The chamfer on edge of thread end shall be 3/64 x 3/64. The large flange diameter shall be 1.120 – 0.002.

e. Guide. Only guides with a chamfer 1/32 x 5/64 inch on end of firing pin bore and 0.075 – 0.015 striker wall thickness shall be reclaimed.

f. Firing pin. Only pins with overall length of 0.743 – 0.030 and striker end length of 0.229 + 0.003 shall be reclaimed.

g. Trunnion. Only trunnions with a 0.206 – 0.014 wall thickness and 1/32 x 1/32 chamfer on inside diameter shall be reclaimed.

3.7 Workmanship. Thrusters shall be free of cracks, splits, or other defects which might prevent proper mating with connecting parts or proper functioning of the device.

3.7.1 Metal defects. All components shall be free from cracks, splits, cold shuts, inclusions, porosity, or any similar defect.

3.7.2 Burr. No part shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.7.3 Foreign matter. No part or assembly shall contain chips, dirt, grease, rust, corrosion, or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements (see 6.2).

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:22Z
Check the source to verify that this is the current version before use.

defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Quality assurance terms and definitions. Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.3 Inspections. Inspections shall be in accordance with MIL-A-2550.

4.2 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed under the following conditions:

a. Temperature: room ambient (68 to 77°F)
b. Altitude: sea level
c. Humidity: 50 percent relative, maximum
d. Atmospheric pressure: 14.7 psi

4.3 Quality conformance inspection.

4.3.1 Lot.

4.3.1.1 Submission of product. Unless otherwise specified by the contracting activity, formation and identification of lots shall be in accordance with MIL-STD-105.

4.3.1.2 Size of lot. For the purpose of identification and shipment, the shipping lots of metal parts assemblies shall consist of 1200 assemblies, maximum, provided the homogeneity requirements of inspection lots have been retained.

4.3.2 Examination. On reprocessed and new parts, one hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects. Four power (4x) magnification may be used in performing visual examination for such defects as burr and foreign matter. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. When compliance with the applicable requirement is in doubt as a result of visual examination, the characteristic shall be measured or gaged to determine acceptability. All nonconforming material shall be rejected.

4.3.2.1 Examination of reclaimed parts. Examination of reusable parts, which require no reprocessing, shall be limited to inspection of defects that require visual type inspection. One hundred percent examination shall be performed for all such defects. All nonconforming parts shall be reprocessed or rejected. For the piston and plug assembly, Drawing 8593866, one hundred percent inspection shall be made for all defects listed in 4.3.2.8 requiring gage inspection.

4.3.2.2 Key, Drawing 8593857, covering a detail of Drawing 8593855.

Characteristics	Defect	Method of Inspection
Critical		

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:22Z
Check the source to verify that this is the current version before use.

None defined

Major

101	Metal defective	Visual
102	Protective finish inadequate or defective	Visual
103	Burr or sharp corner	Visual
104	Foreign matter	Visual

Minor
None defined

4.3.2.3 Body Assembly, Drawing 8593859, covering a detail of Drawing 8593855.

Critical
None defined

Major

101	Pellet missing, loose, or improperly inserted	Manual/Visual
102	Metal defective	Visual
103	Burr	Visual
104	Foreign matter	Visual
105	Protective finish inadequate or defective	Visual
106	Thread damaged or not full	Visual
107	Chamfer on outer edge of threaded end	Visual
108	Chamfer on inner edge of threaded end	Visual
109	Location of pellet	Gage
110	Length from thread undercut to flange	Gage
111	Pitch diameter of large external thread, min	Gage

Minor
None defined

4.3.2.4 Trunnion, Drawing 8593860, covering a detail of Drawing 8593855.

Critical
None defined

Major

101	Metal defective	Visual
102	Protective finish inadequate or defective	Visual
103	Wall thickness	Gage
104	Chamfer on inside diameter	Visual

Minor

201	Burr or sharp corner	Visual
202	Foreign matter	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:22Z
Check the source to verify that this is the current version before use.

4.3.2.5 Retainer, Cartridge, Assembly, Drawing 8593862, covering a detail of Drawing 8593855 (after inserting pellet).

Critical
None defined

Major

101	Location of pellet	Gage
102	Flange diameter	Gage
103	Pitch diameter of large external thread, min	Gage
104	Pellet missing, loose, or improperly inserted	Manual/Visual
105	Chamfer at thread end	Visual
106	Metal defective	Visual
107	Burr	Visual
108	Foreign matter	Visual
109	Protective finish inadequate or defective	Visual
110	Thread damaged or not full	Visual

Minor
None defined

4.3.2.6 Guide, Firing Pin, Drawing 8593863, covering a detail of Drawing 8593855.

Critical
None defined

Major

101	Metal defective	Visual
102	Protective finish inadequate or defective	Visual
103	Burr or sharp corner	Visual
104	Foreign matter	Visual
105	Chamfer at end of firing pin bore	Visual

Minor
None defined

4.3.2.7 Pin, Firing, Drawing 8593864, covering a detail of Drawing 8593855.

Critical
None defined

Major

101	Length from body face to tip of striker	Gage
102	Total length	Gage
103	Metal defective	Visual
104	Protective finish inadequate or defective	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:22Z
Check the source to verify that this is the current version before use.

105	Burr or sharp corner	Visual
106	Foreign matter	Visual

Minor
None defined

4.3.2.8 Piston and Plug Assembly, Drawing 8593866 (Piston 8593867 and Piston Plug 8593868), covering a detail of Drawing 8593855.

Critical
None defined

Major

101	Length from plug face to O-ring groove	Gage
102	Length from small end to end of key escape groove	Gage
103	Metal defective	Visual
104	Protective finish inadequate or defective	Visual
105	Burr	Visual
106	Foreign matter	Visual

Minor
None defined

4.3.2.9 Sleeve, End, Drawing 8593869, covering a detail of Drawing 8593855.

Critical
None defined

Major

101	Chamfer at large end	Gage
102	Width of face at large end	Gage
103	Pitch diameter of large external thread, min	Gage
104	Fillet at shoulder on body outside diameter	Visual
105	Threads damaged or not full	Visual
106	Metal defective	Visual
107	Protective finish inadequate or defective	Visual
108	Burr or sharp corner	Visual
109	Foreign matter	Visual

Minor
None defined

4.3.2.10 Head, Drawing 8593870, covering a detail of Drawing 8593855.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:22Z
Check the source to verify that this is the current version before use.

Critical		
1	Diameter of hole between midsection and small end	Gage
Major		
101	Metal defective	Visual
102	Protective finish inadequate or defective	Visual
103	Burr	Visual
104	Foreign matter	Visual
105	Thread damaged or not full	Visual
Minor		
None defined		

4.3.2.11 Assembly, Drawing 8593855, less all components except Drawings 8593864, 8593863, and MS28775-012.

Critical		
1	Firing pin protrusion	Gage
Major		
101	Any part missing	Visual
102	Protective finish inadequate or defective	Visual
103	Burr	Visual
104	Foreign matter	Visual
Minor		
None defined		

4.3.2.12 Metal Parts Assembly, Drawing 8593855.

Critical		
None defined		
Major		
101	Trunnion not free to rotate 360 degrees	Manual
102	End sleeve not free to rotate 360 degrees clockwise	Manual
103	Any part missing, except as otherwise classified	Visual
104	Improper assembly	Visual
105	Thread damaged or not full	Visual
106	Marking missing, incorrect, or illegible	
107	Burr	Visual
108	Foreign matter	Visual
Minor		
201	Shipping cap or plug missing	Visual
202	Protective finish inadequate or defective	Visual

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:22Z
Check the source to verify that this is the current version before use.

4.3.2.13 Pin, Shear, Drawing 8593284, covering a detail of Drawing 8593855.

Critical
None defined

Major

101	Diameter, max	Gage	
102	Metal defective	Visual	
103	Surface finish improper	Visual	
104	Protective finish inadequate or defective		Visual
105	Burr	Visual	
106	Foreign matter	Visual	

Minor

201	Total length	Gage	
202	Profile of end	Visual	

4.3.2.14 Spring, Piston Locking, Drawing 8593871, covering a detail of Drawing 8593855.

Critical
None defined

Major

101	Load at compressed length	Gage

Minor
None defined

4.3.2.15 Box, Fiberboard (Inner), Packing, Ammunition, Drawing 8797631.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Manual	
202	Box damaged	Visual	
203	Marking missing, incorrect, or illegible		Visual
204	Cut, damaged, or improperly sealed bag		Visual

4.3.2.16 Box, Packing, Ammunition, Drawing 8797632.

Critical
None defined

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:22Z
Check the source to verify that this is the current version before use.

Major
None defined

Minor

201	Contents loose	Visual/Manual
202	Strapping missing, loose, broken, or improperly assembled	Visual/Manual
203	Contents of box exposed	Visual
204	Box damaged	Visual
205	Marking missing, incorrect, or illegible	Visual
206	Number of units incorrect	Visual

4.3.2.17 Box, Fiberboard (Outer), Alternative, Packing, Ammunition, Drawing 9221261.

Critical
None defined

Major
None defined

Minor

201	Contents loose	Manual/Visual
202	Hardware missing, loose, broken, or improperly engaged	Visual
203	Strapping missing, loosed, broken, or improperly engaged	Visual
204	Contents of box exposed	Visual
205	Box damaged	Visual
206	Marking missing, incorrect, or illegible	Visual

4.4 Inspection testing. The following tests shall be performed in accordance with 4.5.

4.4.1 Hydrostatic pressure. Each body and head shall be subjected to the hydrostatic pressure test. Failure of any unit to comply with 3.2 shall be cause for rejection of the subassembly.

4.4.2 Protective finish. The number of units for protective finish tests shall be in accordance with the applicable specification. Failure of any unit in the inspection sample to comply with the requirements of the applicable specification shall be cause for rejection of the lot.

4.4.3 Residual magnetism. Each assembly shall be subjected to the test for excessive residual magnetism as specified in 4.5.3. Failure of any unit in the inspection sample to comply with 3.4 shall be cause for rejection of the assembly.

4.4.4 Shear pin strength. Fifty pins from each lot of shear pins shall be tested for shear strength. Failure of any shear pin to comply with 3.5 shall be cause for rejection of the lot.

4.5 Test methods and procedures.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:22Z
Check the source to verify that this is the current version before use.

4.5.1 Hydrostatic pressure. Prior to application of protective finish the body and head shall be mounted in a device, using a fixture for properly sealing all openings, and subjected to the internal hydrostatic pressure to determine compliance with 3.2.

4.5.2 Protective finish. Test procedures for protective finish shall be in accordance with the applicable specification referenced on the applicable drawing.

4.5.3 Residual magnetism. The assembly shall be tested for excessive residual magnetism using an approved compass in an area free of local magnetic effects by placing the assembly 5 inches from the end and at the same height as the compass, in a north-south horizontal position (head end at south) with the compass set in an east-west heading. The assembly shall be moved at a speed no greater than 10 feet per minute past the compass for its entire length, rotated 90 degrees in the horizontal position about its longitudinal axis and moved slowly back past the compass. This procedure shall be repeated for each 90 degree rotation of the assembly for four full passes.

4.5.4 Shear pin strength. After application of protective finish, the pin shall be subjected to the shear pin strength test to determine compliance with 3.5. The test shall be performed in a fixture in conjunction with a tensile testing machine.

4.6 Inspection of packaging. The sampling and inspection of the packaging, packing, and container marking shall be in accordance with the requirements of section 5.

5. PACKAGING

5.1 Packing and marking. Packing and marking shall be Level A or B, as specified by the contracting activity (see 6.2). Level A shall be in accordance with Drawings 8797631 and 8797632. Level B shall be in accordance with Drawings 8797631 and 9221261.

5.2 Air Force packaging. Air Force packaging shall be in accordance with TPO 00-891-5488.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful but is not mandatory.)

6.1 Intended use. The thruster assembly is used in conjunction with other propellant actuated devices in Aircrew Emergency Escape Systems.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification

b. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2)

c. The responsible inspection activity (see 4.1)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:22Z
Check the source to verify that this is the current version before use.

d. Any special instruction requirements (see 4.2)

e. Level of packing and marking (see 5.1)

6.3 Definition.

6.3.1 Mechanical failure. Mechanical failure is any deformation or breakage of a part, the occurrence of which is other than a design function.

6.4 Exterior components. Remanufactured exterior components are: body, head, trunnion, and end sleeve.

6.5 Residual magnetism. When remanufacturing and loading of assembly is performed by the same facility, the requirement of 3.4 is waived and the residual magnetism test is to be performed on the loaded assembly only.

6.6 Nylon pellets. Components which contain nylon pellets and are to be replated shall have the pellets removed and new pellets installed after plating and in accordance with the applicable drawing.

6.7 Old engraved or stamped lot and serial numbers. Old serial numbers and lot numbers engraved or stamped on metal parts shall be over-stamped by x's to obliterate those markings before application of final protective finish.

6.8 Subject term (keyword) listing.

Ejection systems, aircrew
Emergency escape systems, aircrew
Escape systems, aircrew
Life-saving systems, aircrew

6.9 Changes from previous issue. Marginal notations are not used to identify changes with respect to the previous issue because of the extensiveness of the changes.

Custodian:

Preparing activity
Navy - OS

Navy - OS

Air Force - 99

(Project 1377-0E14)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:22Z
Check the source to verify that this is the current version before use.

Review activity:
Air Force - 70

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T20:22Z
Check the source to verify that this is the current version before use.